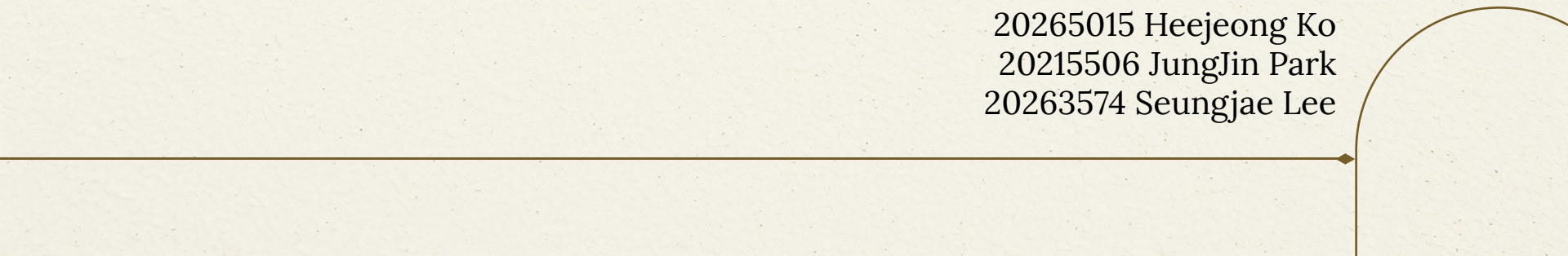




[GCT555 3DID]

# ProxiWall: Distance-Driven Semantic Zoom for HMD-Free Virtual Museum Exploration on Wall Display

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# Approach

01 Recap

- **Environment setting**

- a. **Input** Multiple RGB-D Camera (Front view + Top view)
- b. **Output** 3D Rendered Video — LED Wall (w/o HMD)

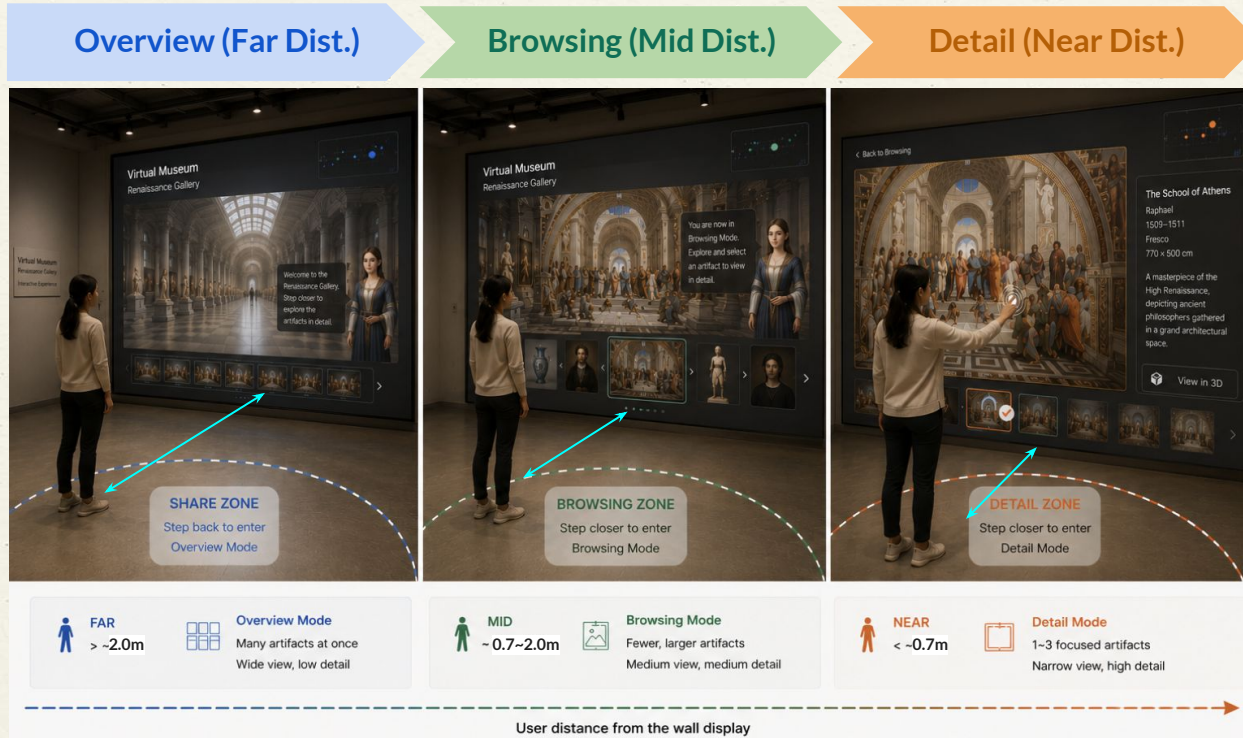
- **Interaction feature**

- a. **Distance-based Semantic Zoom** : User distance from the wall automatically changes the view mode
  - i. Distance controls Overview → Browsing → Detail transition
- b. **Body-based Gallery Navigation**: Lateral movement controls horizontal exploration
- c. **Pseudo touch-based selection** : Touch confirms the final artifact selection

# Distance-based Exploration

## 01 Recap

User distance from the wall automatically changes the view mode.

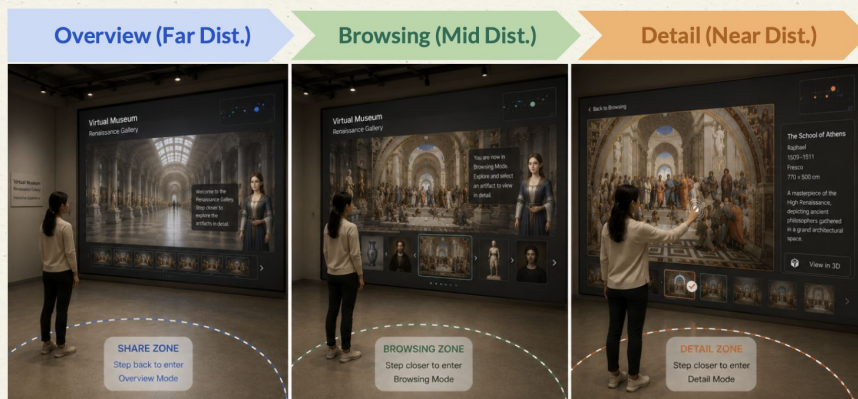




# Interaction Design Parameter

01 Recap

Mapping Walking Distance for Overview ↔ Browsing ↔ Detail switching.



Detail  $\leq 0.7\text{ m}$

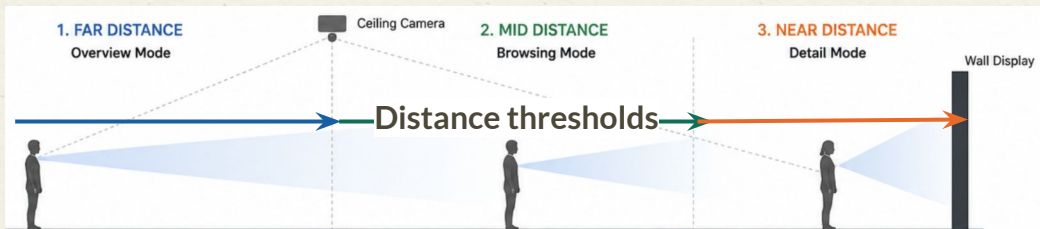
- Selection mode
- Based on comfortable arm reach for direct wall touch

Browsing  $0.7\text{--}2.0\text{ m}$

- Allows users to compare several nearby artifacts

Overview  $> 2.0\text{ m}$

- Roughly two steps back from the wall
- Supports gallery-level scanning and spatial overview



Far  $> 2.0\text{ m}$  | Mid  $0.7\text{--}2.0\text{ m}$  | Near  $\leq 0.7\text{ m}$  | Wall

- These thresholds are design parameters based on bodily reach, walking distance, and visual coverage.

# Core Novelty

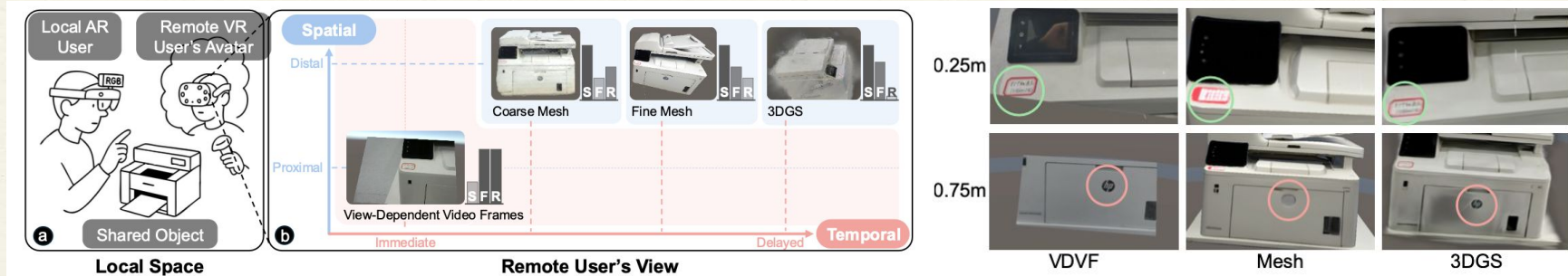
02 Survey

## Beyond HeadZoom: Room-Scale Semantic Zoom on an LED Wall

- **Ceiling camera-based tracking**  
ProxiWall allows visitors to move naturally without wearing an HMD or holding a controller.
- **Larger-scale movement range**  
While HeadZoom uses small head motions, ProxiWall uses full-body walking distance and lateral movement within the exhibition space.
- **LED wall as a shared display**  
Instead of a private HMD view, ProxiWall presents the experience on a large wall display that can be seen by nearby visitors.
- **Semantic zoom, not just visual zoom**  
ProxiWall does not simply magnify the view. It changes the information level from **Overview** → **Browsing** → **Detail** based on the visitor's distance.

# Comparison 1

## 02 Survey



## CollabVisAdapt (IEEE VR 2026)

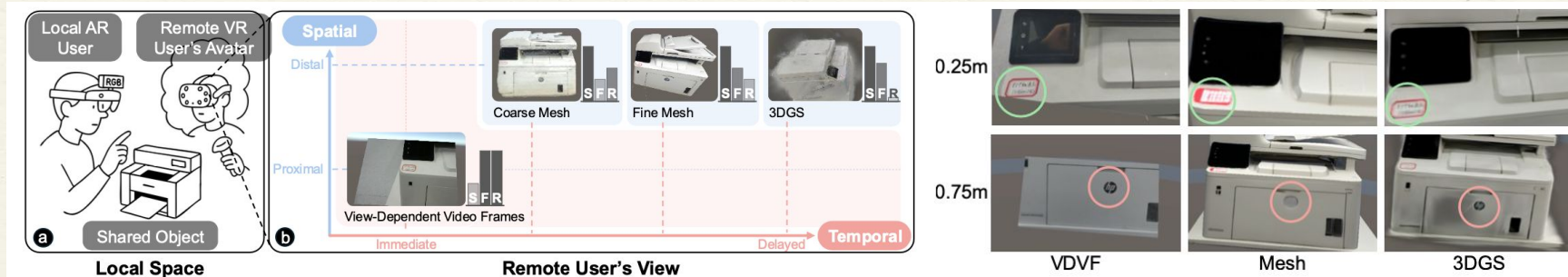
### Interaction Element

- Proximal viewing distances favored high-fidelity VDFV, while distal distances favored high-spatiality 3DGS/Mesh. The estimated preference-switching threshold was  $\sim 0.56\text{m}$ .
- > This supports our use of **distance-based Overview vs. Detail tiers**. but our exact **0.7m / 2.0m** thresholds still require a **task-specific preference-switching threshold study**



# Comparison 1

## 02 Survey



## CollabVisAdapt (IEEE VR 2026)

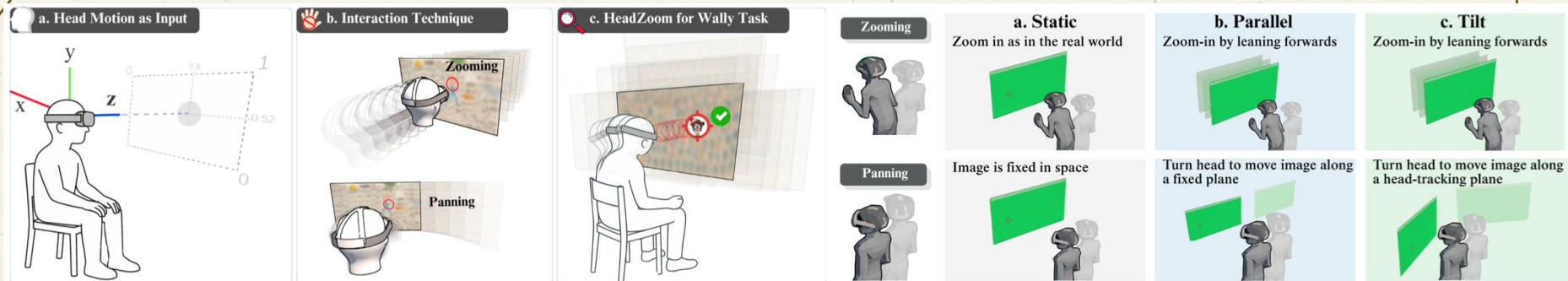
### Study Design

- **Study 1:** Within-subjects design, 4 distances  $\times$  3 modalities
  - Measures: accuracy, observation time, modality preference, optimal viewing distance
  - Psychometric fitting used to identify the **modality switching threshold**
- **Study 2:** Within-subjects design comparing 4 temporal switching schemes (VDVF, MeshS, MeshF, 3DGS)
  - Measures: observation time, error rate, NASA-TLX, preference
- **Final Evaluation:** Adaptive visualization vs. manual switching
  - Measures: task completion time, NASA-TLX, collaboration experience, preference
- -> **Motivates our distance-stratified analysis plan, and binding hypotheses to specific distance bands.**



# Comparison 2

## 02 Survey



## HeadZoom (ISMAR 2025)

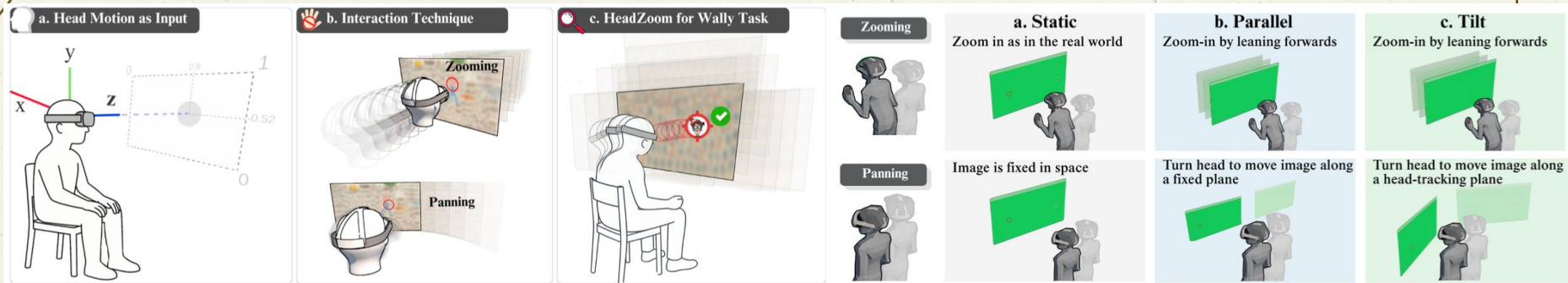
### Interaction Element

- Head Motion as Continuous Input - controls zoom and pan
- Personalized calibration: maps each user's comfortable lean range to interaction range
- Design implication: minimize unnecessary physical effort

-> Our extension: from head-based control to full-body walking-based interaction, borrow the per-user calibration concept for our distance thresholds, and treat 'minimum physical effort' as a design goal.

# Comparison 2

## 02 Survey



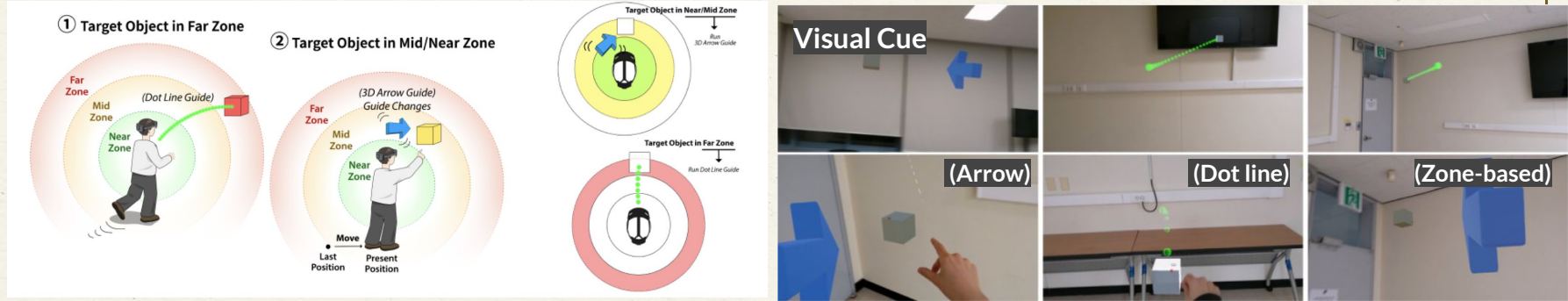
## HeadZoom (ISMAR 2025)

### Study Design

- Within-subjects 3 modes x 2 difficulty (N=31);
  - Where's-Wally targeted-search task;
  - Measurements: completion time, head movement, lean usage, hover time, zoom-change count;
  - post-experience preference ranking.
- > Anchors our Task A targeted-search structure, our per-trial DV set (completion time, target search, mode-dwell)

# Comparison 3

02 Survey



## Distance-Adaptive Guidance (ISMAR-adjunct 2025)

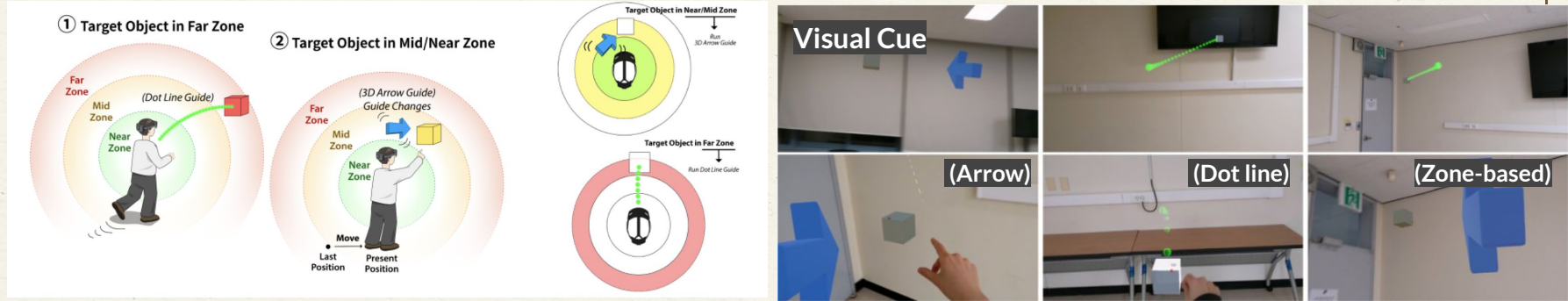
### Interaction Element : Distance-Adaptive Zone Switching

- Divides space into **Near / Mid / Far zones**
- Grounds zone boundaries in **Montello's spatial cognition theory**
- Adapts visual guidance based on user-target distance
- Uses simple directional cues for closer targets and explicit spatial links for farther targets
- Provides a precedent for ProxiWall's **Overview** → **Browsing** → **Detail** interaction tiers



# Comparison 3

02 Survey



## Distance-Adaptive Guidance (ISMAR-adjunct 2025)

### Study Design

- Within-subjects design ( N = 12 participants)
- Compared four guidance conditions: No Guide, 3D Arrow, Dot Line, Zone-Based Guide
- Tested two object-placement layouts: Eye-level placement, Out-of-FoV placement
- Measurement
  1. Spatial awareness / discovery metrics: First Object Discovery Time, Task Completion Time, Mean First-Object Gaze Time
  2. Exploration metrics: Total Movement Distance, Head-Rotation Change
  3. Subjective metrics: NASA-TLX , SUS

-> Provides a model for ProxiWall's per-zone evaluation strategy & Supports combining behavioural, spatial, and subjective measures

# Research Question

## 03 Study Design

### Main RQ:

How can distance-based body input and direct wall touch be designed and evaluated as meaningful interaction modalities for the Overview → Browsing → Selection funnel of a wall-display virtual museum?

- **Sub-RQ 1 – Interaction Usefulness / Task Performance** : Does the calibrated distance-based interaction improve artifact search and selection compared with a baseline wall-display interaction?
- **Sub-RQ 2 – Overall Experience** : How does distance-based body input affect workload, spatial presence, in a wall-display virtual museum?

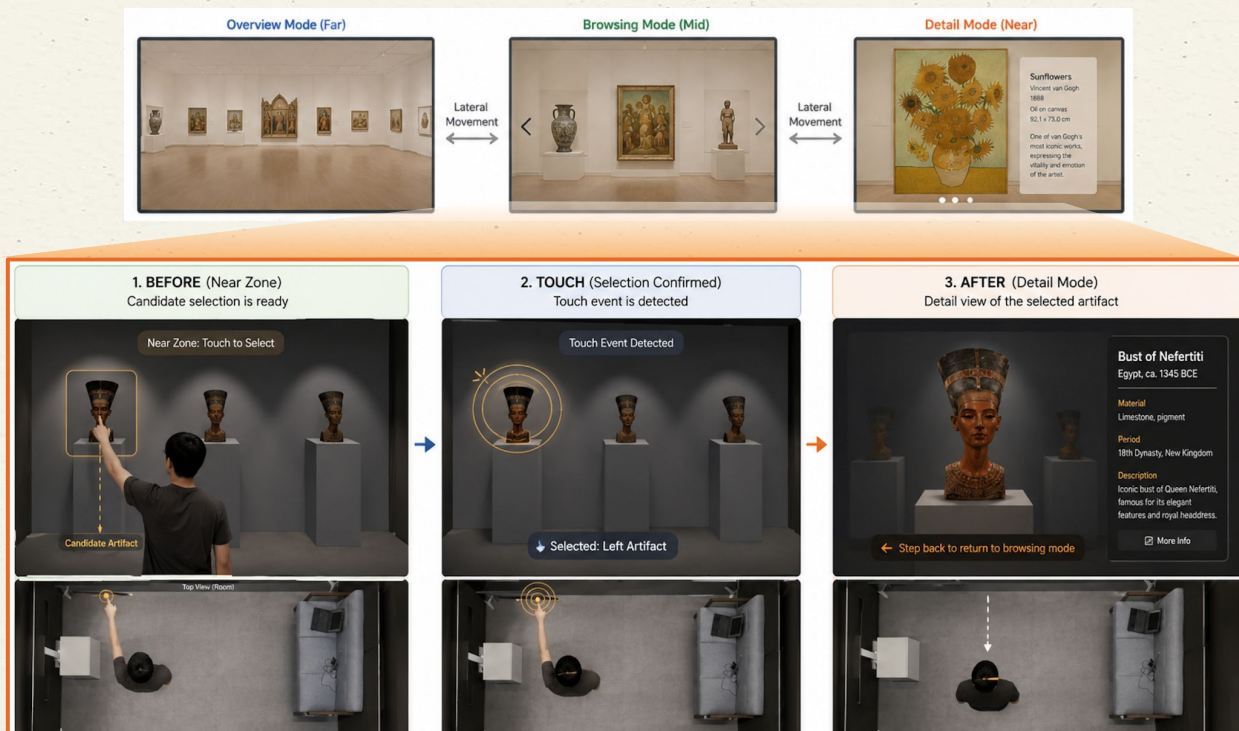
# Hypotheses

- H1** ProxiWall will achieve faster task completion time than Manual Touch Switching.
- H2** ProxiWall will achieve equal or higher task success rate than Manual Touch Switching.
- H3** ProxiWall will reduce perceived mental workload compared to Manual Touch Switching.
- H4** ProxiWall will be perceived as more natural and spatially engaging than Manual Touch Switching.

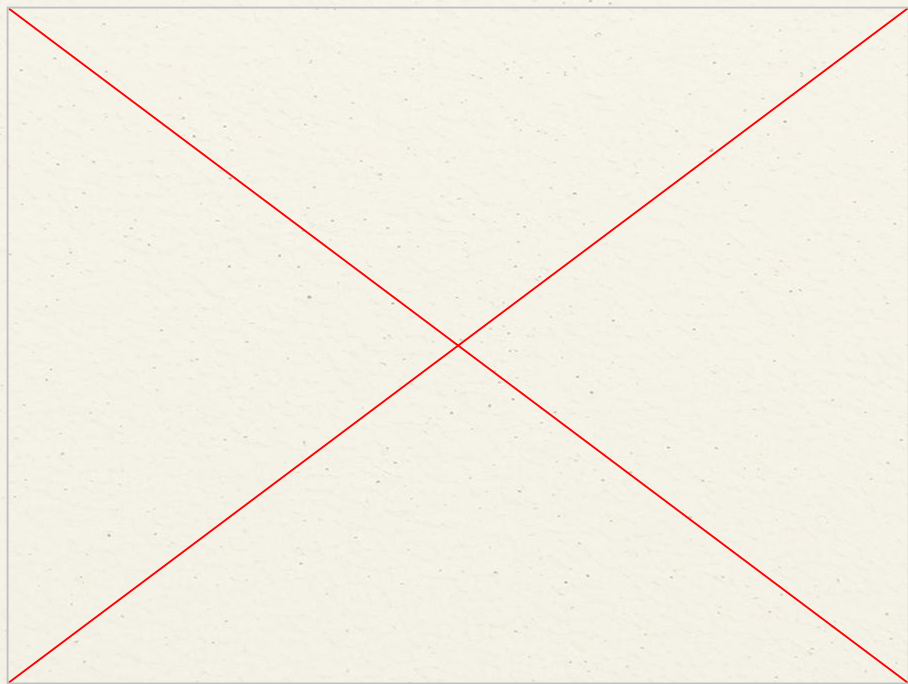


# Study Design

**Goal:** ProxiWall helps users **find and select a specific target artifact** more effectively than baseline.



# Demo.



[https://drive.google.com/file/d/1T3p8zrfwYGBfLxyShyFVRH-53S-Otm3\\_/view?usp=sharing](https://drive.google.com/file/d/1T3p8zrfwYGBfLxyShyFVRH-53S-Otm3_/view?usp=sharing)

# System Setup

## 03 Study Design

| Component      | Specification                                               |
|----------------|-------------------------------------------------------------|
| LED Wall       | 5.4 m × 3.1 m, 1080 × 1920 px                               |
| Ceiling Camera | Azure Kinect (or Webcam), 3.1 m height, captures ~4.7 m FOV |
| Body Tracking  | MediaPipe Pose (33 keypoints, 30 fps)                       |
| Stimuli Pool   | 20 unique 3D-scanned artifacts from 4 periods               |
| Render         | Unity 2026.6000.3 LTS                                       |



# System Setup

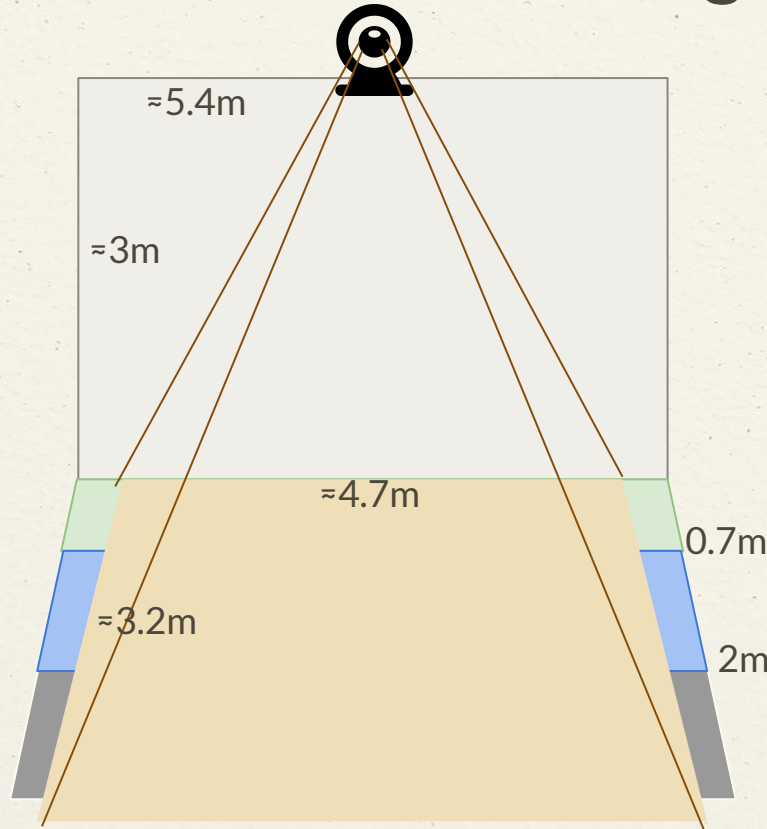
## 03 Study Design

### Ceiling Camera comparison

| Azure Kinect               | Webcam                    |
|----------------------------|---------------------------|
| captures ~4.7 m FOV (120°) | captures ~3.5 m FOV (85°) |



# Environment Setting



-  = Overview Mode
-  = Browsing Mode
-  = Detail Mode
-  = LED Wall
-  = Captured Area

# Condition

Participants : target N=15 / Within-subjects design

## C1. ProxiWall (Distance-Driven) [Ours]

Automatic mode switching by distance + lateral body navigation  
+ touch-to-select

### 1. Mode Switching (by Distance)



### 2. Browsing Navigation (by Lateral Body Movement)



Step left / right  
to browse  
artifacts

### 3. Selection (Touch)



Touch the artifact  
to make selection

VS.

## C2. Manual Touch Switching

Explicit mode switching by on-screen buttons  
+ touch arrows/swipe for navigation + touch-to-select

### 1. Mode Switching (by On-Screen Buttons)



Tap the button to switch modes

### 2. Browsing Navigation (by Touch Arrows / Swipe)



Tap arrows  
or swipe  
to browse

### 3. Selection (Touch)



Touch the artifact  
to make selection



# Measurement

## Objective Measures

| Metric                                        | Measurement                                                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Task Completion Time</b>                   | Time from Task Start to Target Artifact Selection (seconds)                                         |
| <b>Selection Accuracy (Task Success Rate)</b> | Percentage of trials in which the participant correctly selected the target artifact                |
| <b>Selection Error Rate</b>                   | Percentage of trials involving incorrect object selection or failure to select the target artifact. |

## Subjective Measures

| Metric                             | Measurement                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>NASA-TLX</b>                    | Workload, mental demand, and physical demand                                                  |
| <b>Presence (MEC-SPQ, 8 items)</b> | Spatial situation model, self-location, and possible actions in the virtual museum            |
| <b>Perceived Naturalness</b>       | <b>Custom</b> 7-point Likert items measuring whether the interaction felt natural, intuitive. |

## Qualitative Evaluation

|                                  |                                                           |
|----------------------------------|-----------------------------------------------------------|
| <b>Semi-Structured Interview</b> | Naturalness, immersion, discomfort, and <u>preference</u> |
|----------------------------------|-----------------------------------------------------------|

# Procedure

## Study

## Procedure: Targeted Artifact Search

**Task:** Participants are asked to find a specific target artifact in the virtual museum (shown by name, thumbnail, or short description) and select the matching artifact on the wall display.

1

### Target Preview

Target artifact information is shown.

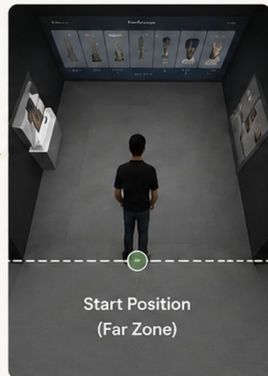


⌚ 3 sec

2

### Return to Start

Participant returns to the start position (far zone).



3

### Search

Participant searches for the target using the assigned interaction condition.



 Move to explore  
(forward/backward, left/right)

 Touch to select  
the artifact

4

### Select Artifact

Participant touches the artifact they believe matches the target.



5

### Feedback

Correct / incorrect feedback is shown.



System logs trial data automatically.

# Timeline

## 04 Timeline

| Description         | Sub-task                   | Week |   |   |    |    |    |    |    |    |    | Owner | Status      |
|---------------------|----------------------------|------|---|---|----|----|----|----|----|----|----|-------|-------------|
|                     |                            | 7    | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |       |             |
| Interaction Concept | Ideation + scenarios       |      |   |   |    |    |    |    |    |    |    | HJ    | Completed   |
| Tracking Setup      | Ceiling RGB-D rig install  |      |   |   |    |    |    |    |    |    |    | SJ    | Completed   |
|                     | Skeleton + pipeline        |      |   |   |    |    |    |    |    |    |    | JJ    | Completed   |
| Unity Prototype     | Scene + asset import       |      |   |   |    |    |    |    |    |    |    | JJ    | Completed   |
|                     | Wall display renderer      |      |   |   |    |    |    |    |    |    |    | JJ    | Completed   |
|                     | Sensor → Unity bridge      |      |   |   |    |    |    |    |    |    |    | JJ    | Completed   |
| Parameter Design    | Distance zone calibration  |      |   |   |    |    |    |    |    |    |    | HJ    | Completed   |
|                     | Hysteresis & dwell tuning  |      |   |   |    |    |    |    |    |    |    | HJ/SJ | Completed   |
|                     | CD-gain function design    |      |   |   |    |    |    |    |    |    |    | SJ    | Completed   |
|                     | Touch size & latency spec  |      |   |   |    |    |    |    |    |    |    | HJ    | Completed   |
| Semantic Zoom       | Overview mode layout       |      |   |   |    |    |    |    |    |    |    | SJ    | Completed   |
|                     | Browsing + focus indicator |      |   |   |    |    |    |    |    |    |    | SJ    | Completed   |
|                     | Detail + metadata panel    |      |   |   |    |    |    |    |    |    |    | SJ    | Completed   |
| FoV Mapping         | Wall geometry calibration  |      |   |   |    |    |    |    |    |    |    | JJ    | In Progress |
|                     | Angular mapping function   |      |   |   |    |    |    |    |    |    |    | JJ    | In Progress |
|                     | Multi-user perspective     |      |   |   |    |    |    |    |    |    |    | JJ    | In Progress |
| User Study          | Pilot (n = 3–5)            |      |   |   |    |    |    |    |    |    |    | SJ    | In Progress |
|                     | Main study (n = 15–20)     |      |   |   |    |    |    |    |    |    |    | SJ    | In Progress |
| Analysis            | Quantitative analysis      |      |   |   |    |    |    |    |    |    |    | HJ    | Not started |
|                     | Qualitative analysis       |      |   |   |    |    |    |    |    |    |    | HJ    | Not started |
|                     | Writing & figure prep      |      |   |   |    |    |    |    |    |    |    | HJ    | Not started |

### OWNERS

■ HJ · Heejeong

■ SJ · Seungjae

■ JJ · Jungjin



# Thanks!

**ProxiWall:Distance-Driven Semantic Zoom for  
HMD-Free Virtual Museum Exploration on Wall  
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